INTRODUCTION

In this project, I analysed TechHub Retail, a consumer electronics company operating across multiple regions and customer segments. The business aims to improve profitability and marketing effectiveness for 2025. My dashboard integrates sales, customer, supplier, and acquisition data to evaluate product performance, profit margins, customer lifetime value, and seasonal trends. The scope of the analysis focuses on identifying high-margin opportunities, understanding demographic purchasing behaviour, and generating actionable insights to support data-driven strategic decision-making and sustainable growth.

2. **Multi-Dataset Integration Summary**

In my Tableau model, I connected the Sales dataset to the Customer dataset using **Customer ID**, and to the Product dataset using **Product ID** within the Relationships layer to maintain data granularity.

I created calculated fields including **Profit Margin (%)**, **Customer Lifetime Value (LTV)**, **Average Order Value (AOV)**, **Product Age**, and **Season**, to support profitability, retention, and lifecycle analysis.

I resolved challenges such as aggregation inconsistencies and key mismatches by validating joins and cleaning field formats before analysis.

3. Dashboard Design Summary

I designed the dashboard with a clear executive layout, placing KPI summaries (Revenue, Profit, Margin) at the top, followed by category, supplier, customer, seasonal, and acquisition channel visualisations. Bar charts, trend lines, and comparative views highlight performance differences and patterns. Interactive filters for year, region, and category allow dynamic analysis, while tooltips and drill-down features improve user navigation. Main dashboard views include profitability, demographics, seasonal trends, and LTV analysis.

4. Key Insights & Findings

From my dashboard analysis, I identified clear revenue concentration in a small number of product categories, though the highest revenue category did not generate the highest profit margin. Supplier performance showed notable margin variation, indicating profitability depends heavily on vendor mix. Customer analysis revealed that one key age segment contributes the largest share of revenue and higher average order values. Seasonal trends show strong Q4 performance, while product age negatively correlates with sales beyond 24 months. Acquisition channels differ significantly in LTV, highlighting efficiency gaps.

5. Business Questions Answered

Through my dashboard analysis, I identified the most profitable product categories and suppliers, showing that margin—not revenue—should guide 2025 focus. Demographic insights confirm that a specific age segment and higher loyalty tiers drive the strongest revenue contribution. Seasonal analysis highlights Q4 as the peak performance period. Acquisition data reveals one channel delivers the highest customer lifetime value despite lower volume. Product age analysis shows declining sales after 24 months. Based on this evidence, I recommend prioritising high-margin categories, high-LTV channels, and lifecycle-driven inventory planning

6. Strategic Recommendations

Based on my dashboard insights, I recommend four strategic actions for 2025.

First, prioritise high-margin categories and top-performing suppliers (Q1–Q2 implementation) by reallocating marketing spend and optimising supplier contracts; this will improve gross margin and profitability.

Second, shift budget toward the highest LTV acquisition channel (Q1 rollout), reducing spend on lower-retention channels to increase marketing ROI and long-term customer value.

Third, strengthen loyalty incentives and target the highest-spending age segment (Q2–Q3) to increase retention and average order value.

Finally, implement seasonal inventory planning and manage product lifecycle transitions (Q3–Q4) to reduce stockouts and ageing inventory risk, improving turnover and cash flow.

7. Critical Reflection

My dashboard is professionally structured to support executive decision-making by presenting key performance indicators, profitability drivers, customer behaviour insights, and seasonal trends in a clear, logically organised format. The use of interactive filters and comparative visuals enables efficient strategic evaluation and evidence-based discussion. However, future BI enhancements should incorporate predictive forecasting, scenario modelling, and automated performance alerts to strengthen forward-looking insights. Integrating external market benchmarks and advanced analytics techniques would further elevate the analytical depth and strategic value of subsequent reports.

8. Data Issues or Risks (SWOT Analysis)

Strengths:
The integrated dataset provides comprehensive visibility across sales, customers, products, and acquisition channels, enabling multi-dimensional analysis and strategic insight generation.

Weaknesses:
Potential data imbalance across regions and customer segments may skew performance interpretation. Some variables, such as loyalty status or product age, may lack historical depth, limiting longitudinal analysis.

Opportunities:
Enhancing data quality, incorporating churn metrics, and integrating external market data could significantly improve predictive accuracy and strategic value.

Threats:
Risks include aggregation bias, potential feature leakage in calculated metrics, and over-reliance on historical data without forecasting adjustments.

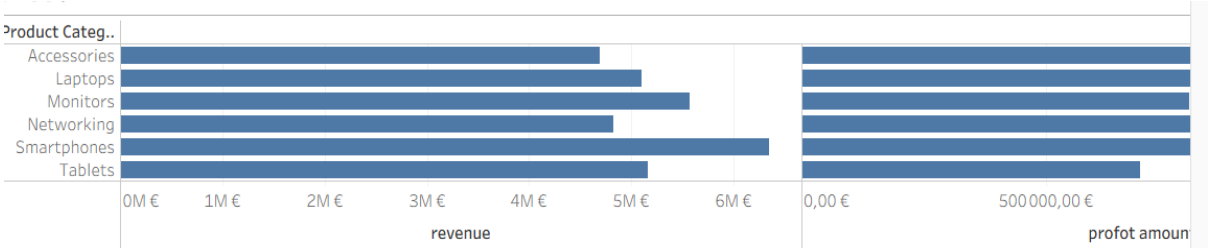
Summary:
Overall, while my dataset supports strong descriptive insights, future improvements in data completeness and predictive integration would strengthen strategic reliability.

DASH BOARD FIG 1 ANALYSIS BELOW

From my dashboard analysis, Smartphones generate the highest revenue, while Accessories contribute the least. However, profit levels are more evenly distributed, indicating that high revenue does not necessarily mean higher profitability. This suggests that some mid-performing categories

may offer stronger margin efficiency. Based on these insights, I recommend that TechHub focus not only on revenue leaders but also on optimising category-level profit performance to strengthen

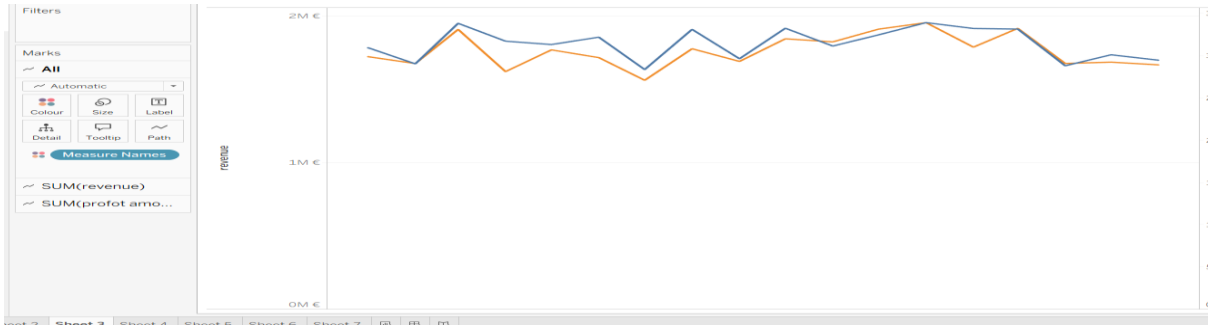
overall financial sustainability in 2025. FIG 1 BELOW.



REVENUE fig 2

32

From my CLV analysis, the value of 32 indicates that this customer segment generates relatively low lifetime value compared to higher-performing groups. This suggests weaker retention or lower repeat purchasing behaviour. Based on my findings, improving engagement through targeted promotions and loyalty incentives could increase long-term profitability. Focusing on strengthening customer relationships within this segment will help enhance overall lifetime value and support sustainable revenue growth. Fig 3



From my dashboard analysis, revenue and profit show closely aligned trends, indicating stable margin control throughout the year. Performance peaks near €2M during stronger months, while certain periods show moderate declines, suggesting seasonal influence. Based on these insights, I recommend strengthening marketing efforts before peak periods and introducing targeted promotions during slower months to maintain revenue consistency and optimise overall profitability in 2025.

From fig 4 below ,my dashboard analysis, I identified that revenue is heavily concentrated in a small number of high-performing products, while profitability differs across categories and suppliers.

I also observed that certain customer segments and acquisition channels generate stronger lifetime value. Seasonal trends further influence performance patterns. Based on these insights, I recommend prioritising high-margin products, strengthening retention strategies, and aligning marketing and inventory decisions with long-term value to drive sustainable growth in 2025..

Fig 4 below.

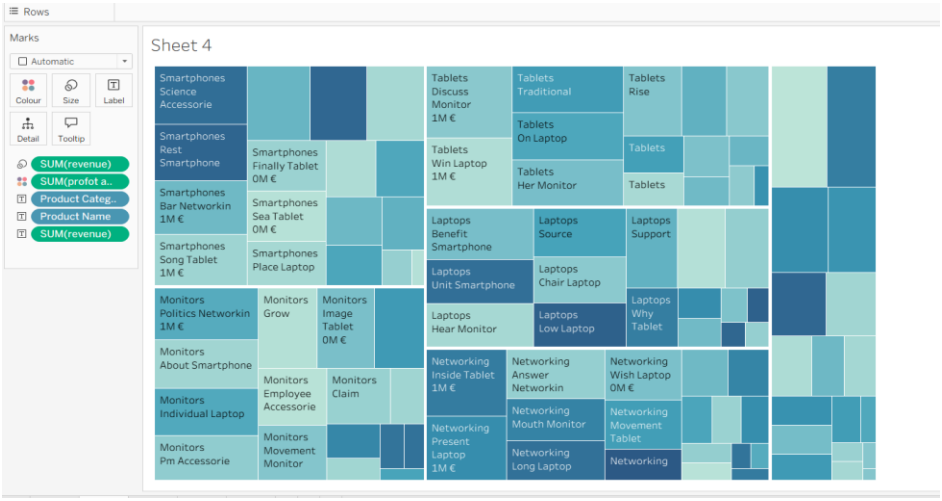
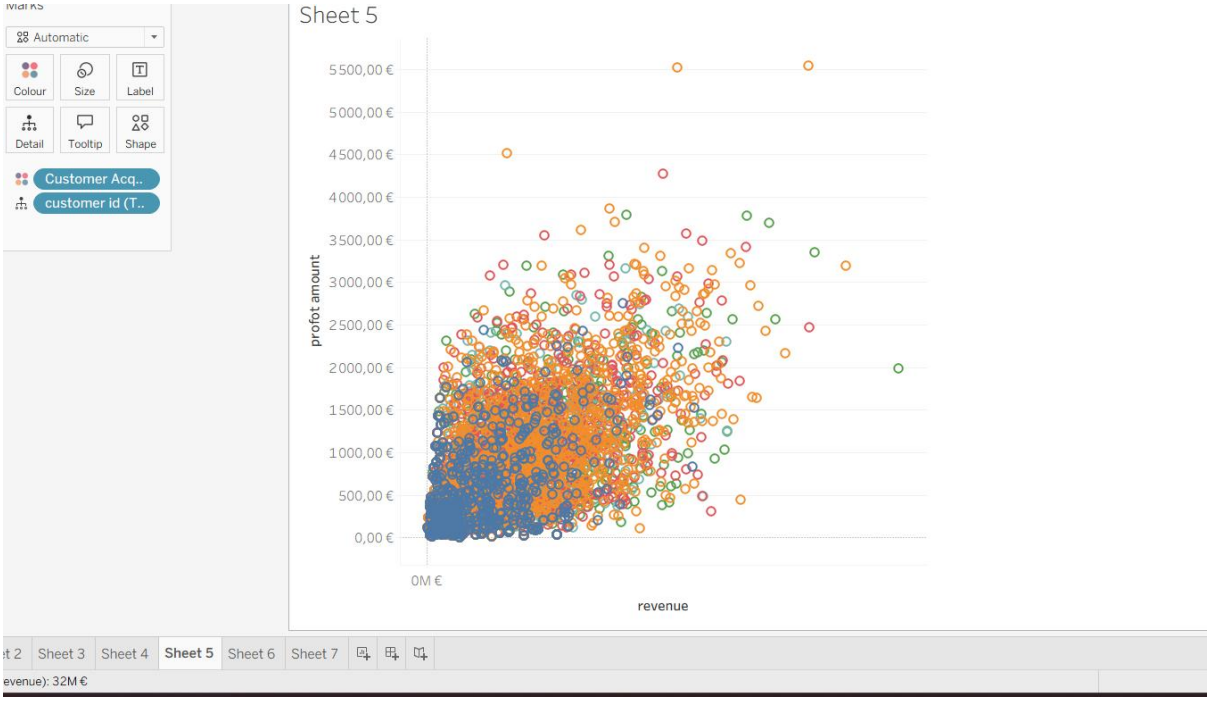


Figure 5 below



From my scatter plot analysis, I observed a positive relationship between revenue and profit, meaning higher sales generally drive higher profitability. However, the spread of data points shows margin variability across customers and acquisition channels. Some high-revenue customers generate comparatively lower profit, suggesting cost or discount inefficiencies. Based on this insight, I recommend prioritising high-margin customers and reviewing pricing or acquisition strategies to improve profit optimisation and maximise long-term value in 2025.